UNIT-4
Diesel And Gas Power Plant

Introduction

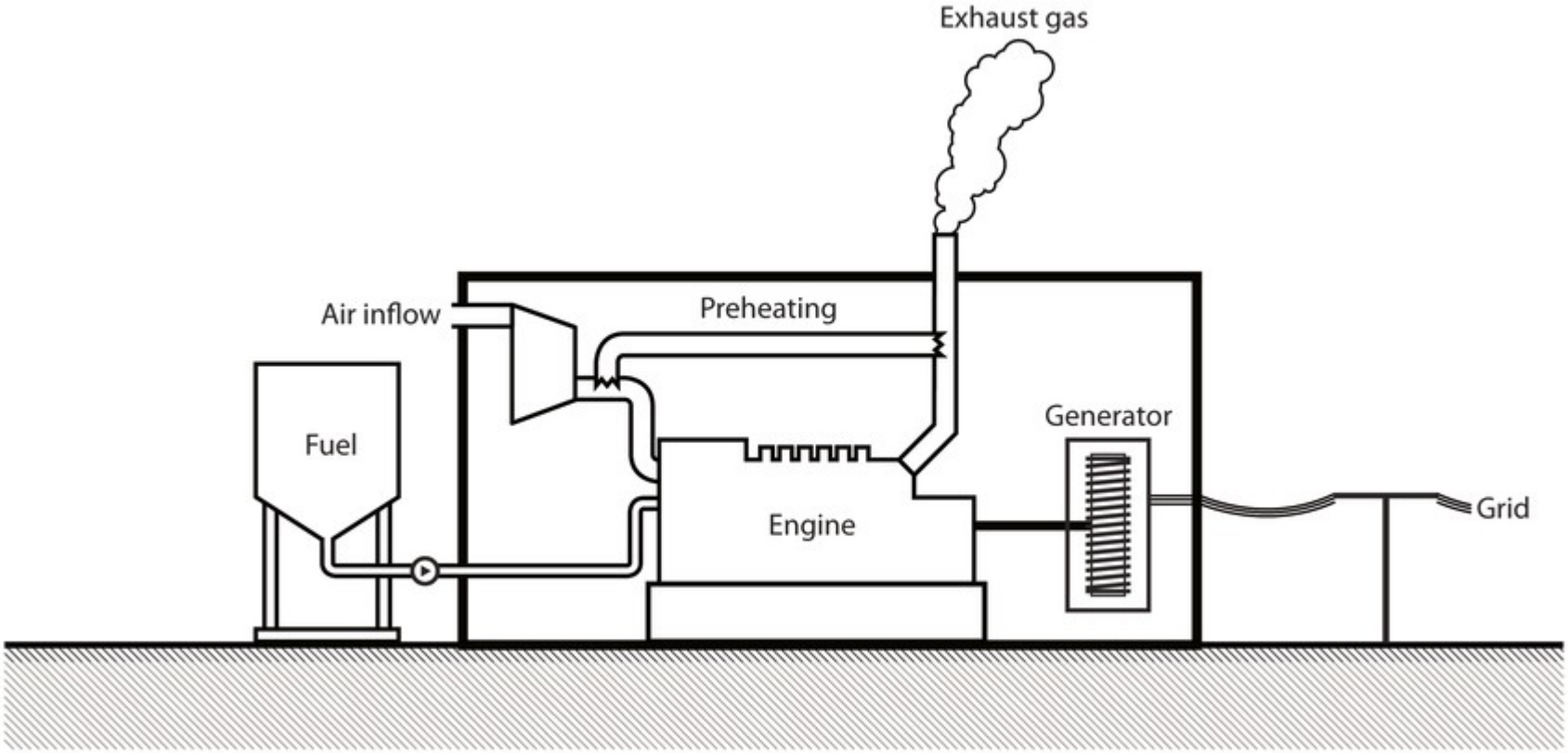
A diesel power plant uses a diesel engine (internal combustion engine) to generate electricity

Converts chemical energy of fuel → mechanical energy → electrical energy

Used for small-scale power generation and backup supply

Working Principle

Fuel Energy→Mechanical Energy→Electrical Energy



Explanation:

- Diesel fuel injected into engine cylinder
- Air compressed → fuel ignites automatically
- Expanding gases rotate crankshaft
- Generator converts rotation → electricity

Types of Diesel Power Plants

1. Based on Engine Speed:

Low-speed engines (<300 rpm)
Medium-speed engines (300–1000 rpm)
High-speed engines (>1000 rpm)

2. Based on Cycle:

Two-stroke engines
Four-stroke engines

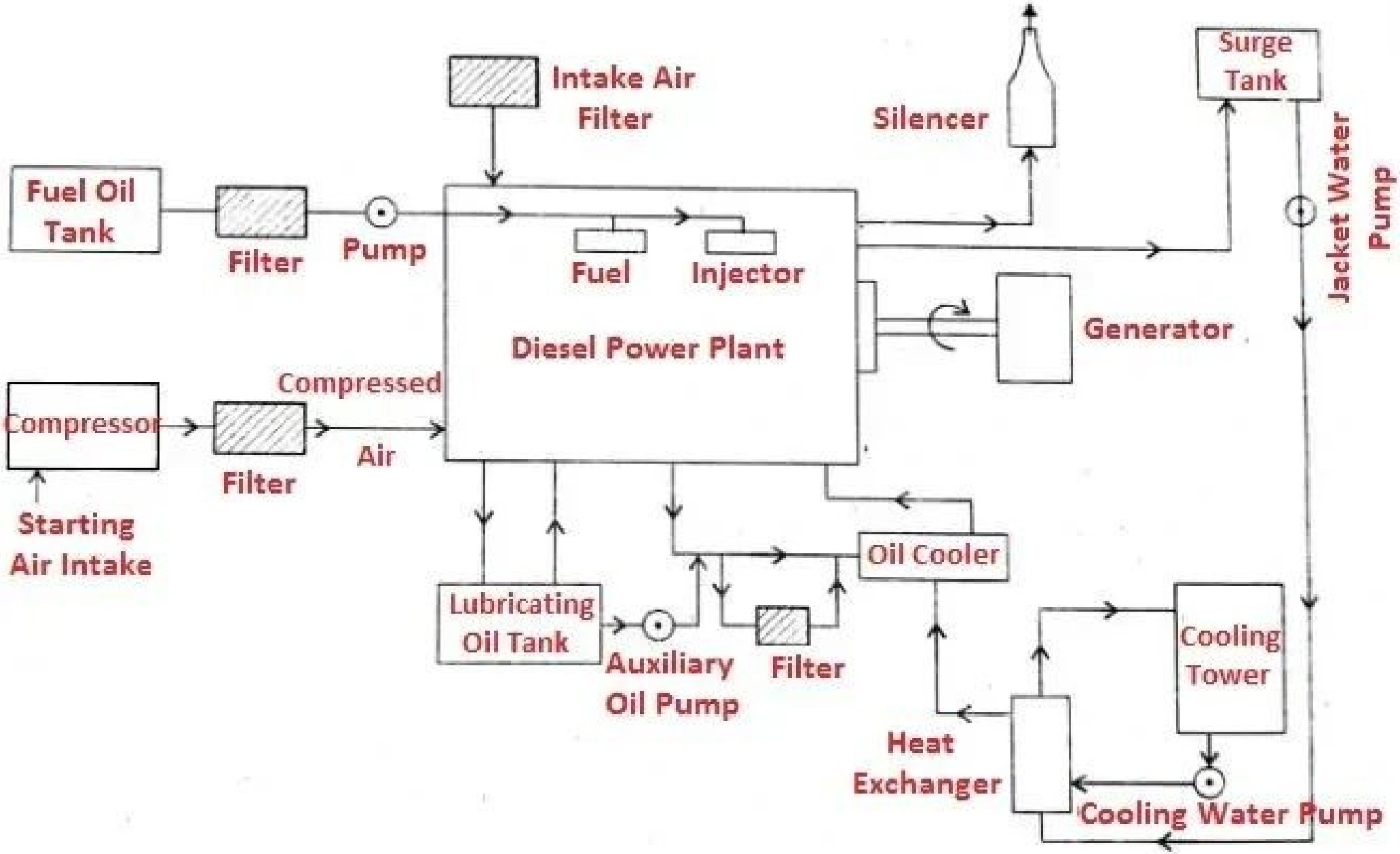
3. Based on Application:

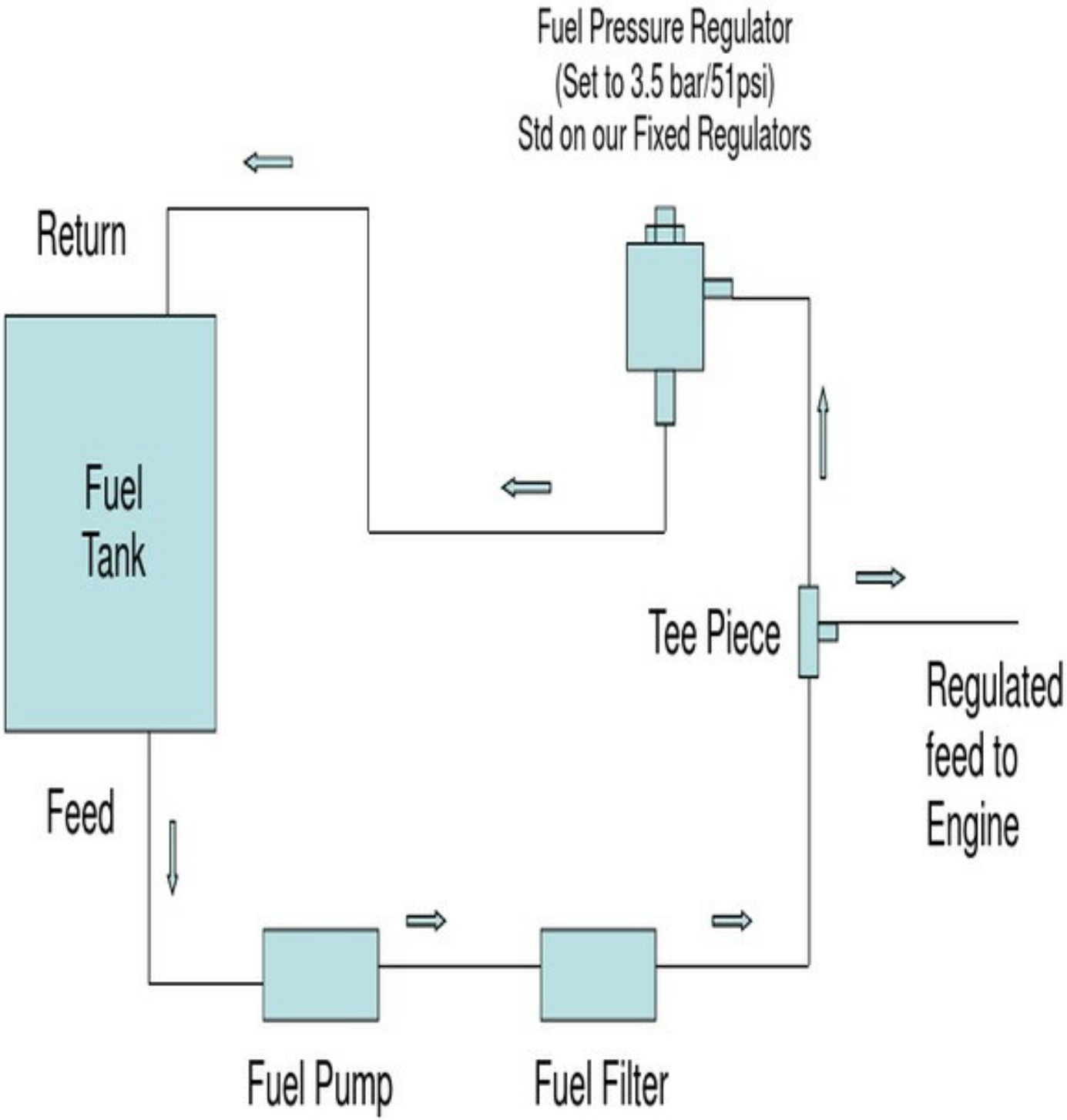
Stationary plants
Mobile/portable plants

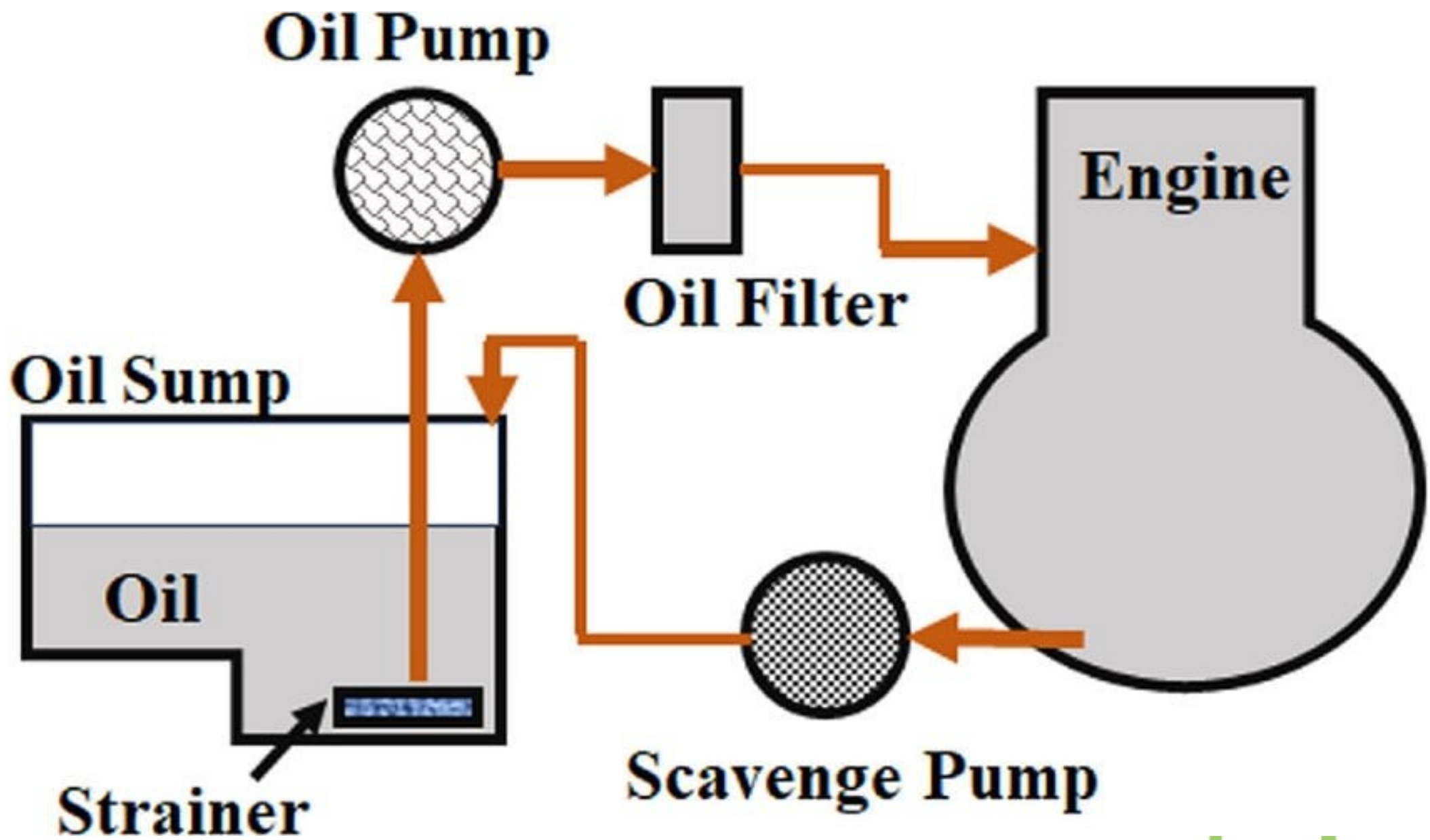
Components & Functions

Main Components:

- Diesel engine – prime mover
- Fuel system – storage, pump, injector
- Air intake system – supplies air for combustion
- Exhaust system – removes gases
- Cooling system – prevents overheating
- Lubrication system – reduces friction
- Starting system – initiates engine
- Generator (alternator) – produces electricity


Intake Air Filter
Silencer
Surge Tank
Jacket Water Pump
Fuel Oil Tank
Filter
Pump
Fuel
Injector
Diesel Power Plant
Generator
Compressed Air
Compressor
Filter
Starting Air Intake
Oil Cooler
Lubricating Oil Tank
Auxiliary Oil Pump
Filter
Cooling Tower
Heat Exchanger
Cooling Water Pump


Fuel Pressure Regulator
(Set to 3.5 bar/51psi)
Std on our Fixed Regulators
Return
Fuel
Tank
Tee Piece
Regulated
feed to
Engine
Feed
Fuel Pump
Fuel Filter


Oil Pump
Oil Filter
Engine
Oil Sump
Oil
Strainer
Scavenge Pump

Selection of Diesel Engine

Factors:

- Power requirement
- Speed of operation
- Fuel efficiency
- Cost and maintenance
- Load variation

Selection Guide:

Small load → high-speed engine

Continuous heavy load → low-speed engine

Applications of Diesel Power Plant

Emergency/backup power supply
Small industries
Remote and rural areas
Peak load plants
Standby generators in hospitals, offices

Advantages
Quick starting
Low initial cost
Simple design
Less water requirement

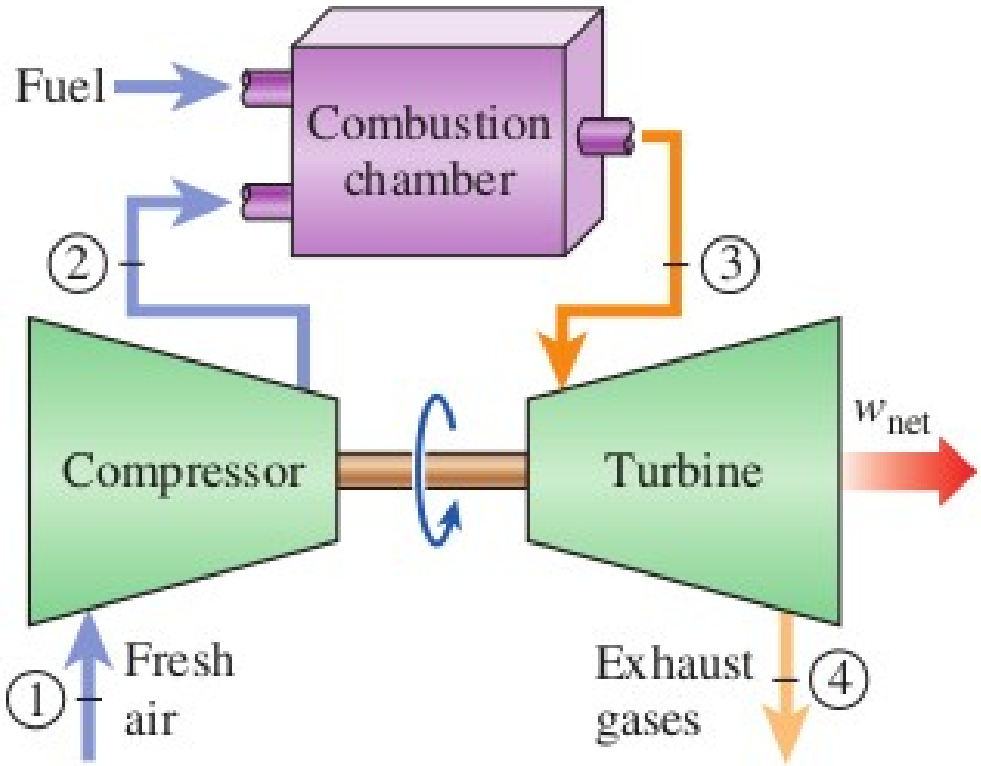
Disadvantage
High fuel cost
Noise and pollution
Limited capacity
Higher maintenance

Introduction to Gas Power Cycle

The Brayton cycle was first proposed by George Brayton for use in the reciprocating oil-burning engine that he developed around 1870.

Today, it is used for gas turbines only where both the compression and expansion processes take place in rotating machinery.

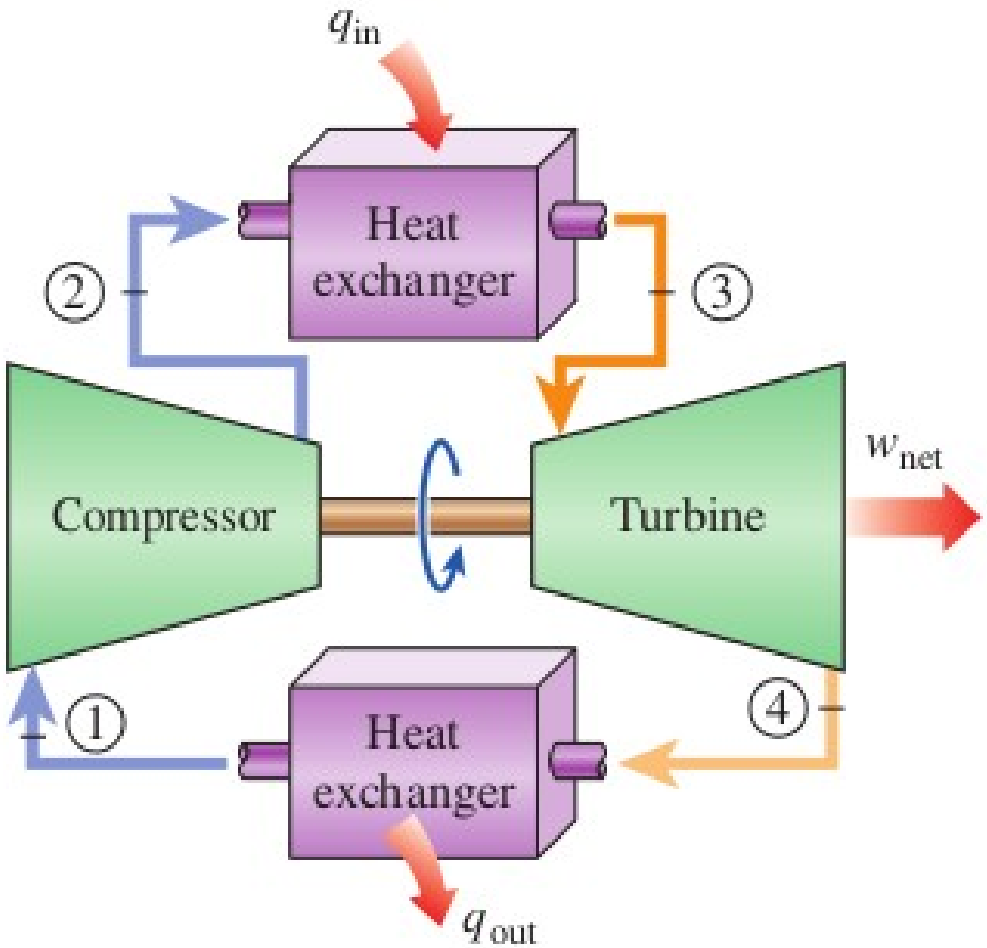
Gas turbines usually operate on an open cycle.



An open-cycle gas-turbine engine.

Introduction to Gas Power Cycle

The open gas-turbine cycle just described can be modeled as a closed cycle, by using the air-standard assumptions.



Introduction to Gas Power Cycle

Air Standard Assumption:

1. The working fluid is air, which continuously circulates in a closed loop and always behaves as an ideal gas.
2. All the processes that make up the cycle are internally reversible.
3. The combustion process is replaced by a heat-addition process from an external source
4. The exhaust process is replaced by a heat-rejection process that restores the working fluid to its initial state.
5. Another assumption used to simplify the analysis is that air has constant specific heats whose values are determined at room temperature.

Brayton Cycle

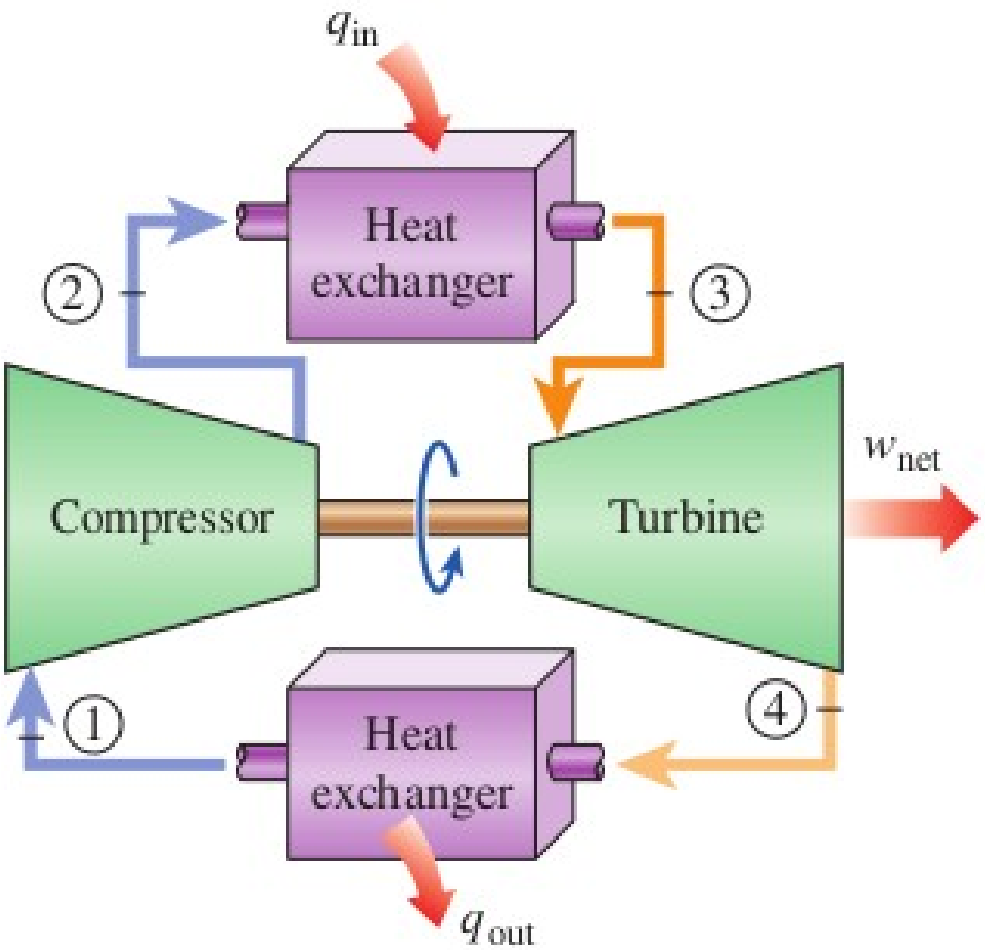
The ideal cycle that the working fluid undergoes in this closed loop is the Brayton cycle, which is made up of four internally reversible processes:

1-2 Isentropic compression (in a compressor)

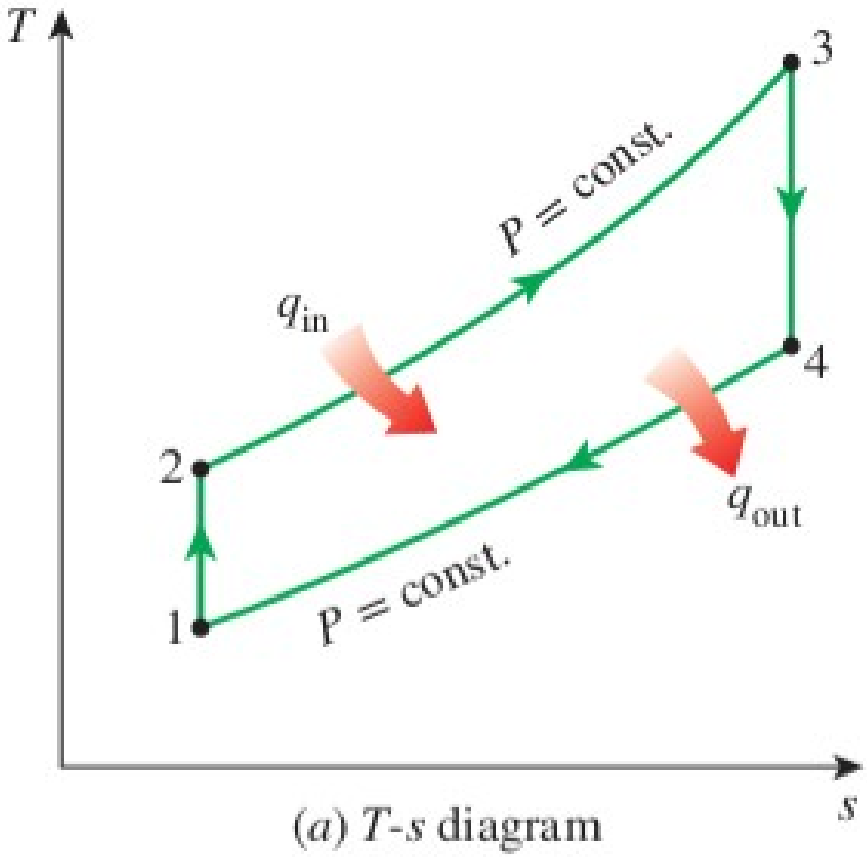
2-3 Constant-pressure heat addition

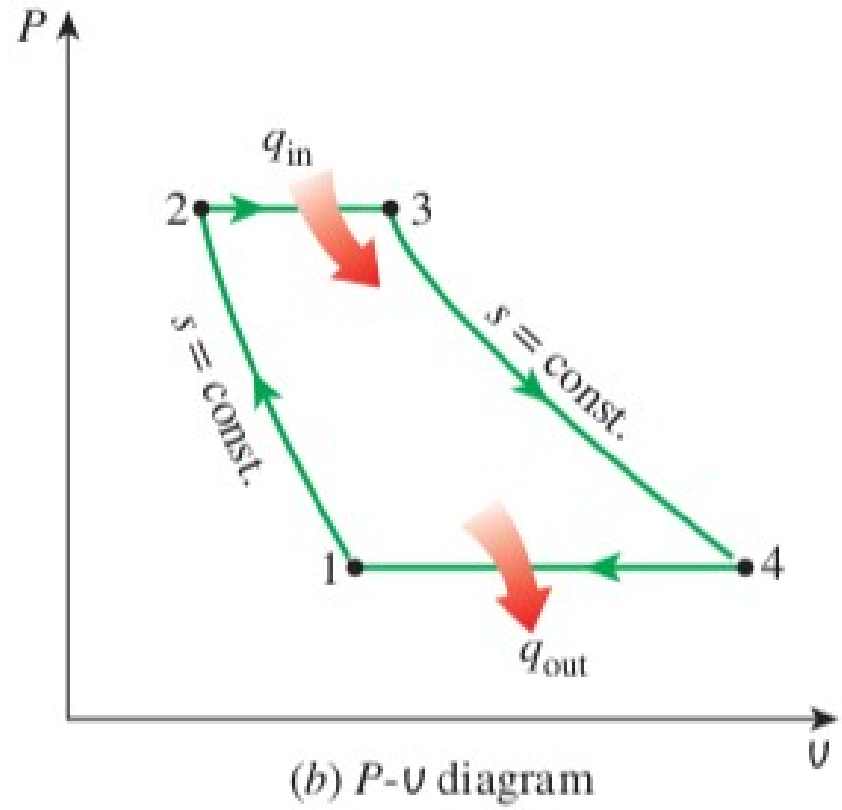
3-4 Isentropic expansion (in a turbine)

4-1 Constant-pressure heat rejection



Brayton Cycle on PV and TS diagram





T-S and P-V diagram for Ideal Brayton Cycle

Brayton Cycle Efficiency

$$\eta = \frac{Net\ Work\ done}{Heat\ Input} = \frac{W_{turbine} - W_{comp}}{Q_{input}}$$

$$W_{turbine} = \dot{m} . C_p (T_3 - T_4);$$

$$BWR = \frac{W_{comp}}{W_{turbine}};$$

$$W_{comp} = \dot{m} . C_p (T_2 - T_1);$$

$$Q_{input} = \dot{m} . C_p (T_3 - T_2);$$

$$\eta = \frac{\dot{m} . C_p (T_3 - T_4) - \dot{m} . C_p (T_2 - T_1)}{\dot{m} . C_p (T_3 - T_2)}$$

$$\eta = \frac{(T_3 - T_4) - (T_2 - T_1)}{(T_3 - T_2)} = \frac{(T_3 - T_2) - (T_4 - T_1)}{(T_3 - T_2)}$$

Brayton Cycle Efficiency

$$\eta = 1 - \frac{(T_4 - T_1)}{(T_3 - T_2)}$$

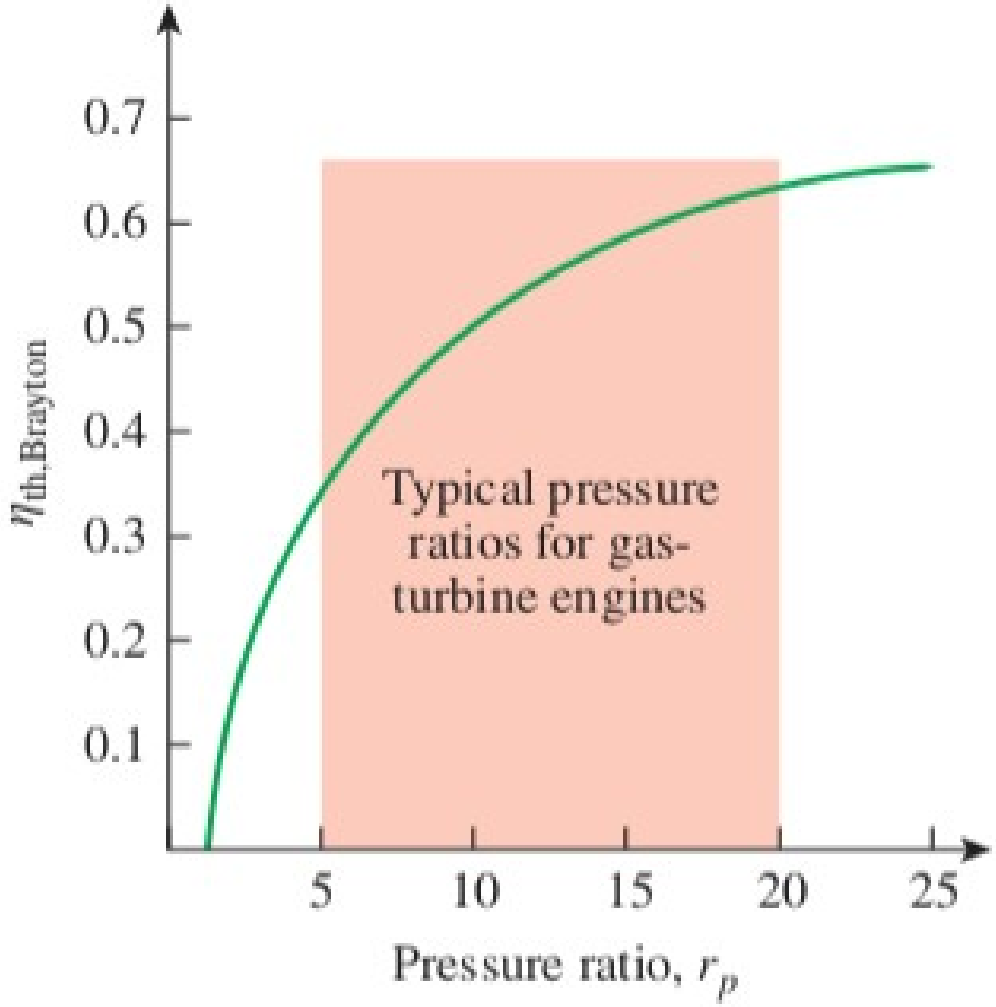
$$T_2 = T_1 r_p^{\frac{\gamma-1}{\gamma}} \qquad T_3 = T_4 r_p^{\frac{\gamma-1}{\gamma}} \qquad r_p = \frac{P_2}{P_1}$$

$$\eta = 1 - \frac{(T_4 - T_1)}{T_4 r_p^{\frac{\gamma-1}{\gamma}} - T_1 r_p^{\frac{\gamma-1}{\gamma}}} = 1 - \frac{(T_4 - T_1)}{r_p^{\frac{\gamma-1}{\gamma}} (T_4 - T_1)}$$

$$\eta = 1 - \frac{1}{r_p^{\frac{\gamma-1}{\gamma}}}$$

Brayton Cycle Efficiency

$$\eta = 1 - \frac{1}{r_p^{\frac{\gamma-1}{\gamma}}}$$



Brayton Cycle Problem

A gas-turbine power plant operating on an ideal Brayton cycle has a pressure ratio of 8. The gas temperature is 300 K at the compressor inlet and 1300 K at the turbine inlet. Using the air-standard assumptions, determine

(a) the gas temperature at the exits of the compressor and the turbine,

(b) the back work ratio, and

(c) the thermal efficiency.

(d) Heat Input in cycle

$$\eta = 1 - \frac{1}{r_p^{\frac{\gamma-1}{\gamma}}}$$

$$W_{turbine} = \dot{m} . C_p (T_3 - T_4);$$

$$W_{comp} = \dot{m} . C_p (T_2 - T_1);$$

$$Q_{input} = \dot{m} . C_p (T_3 - T_2);$$

$$BWR = \frac{W_{comp}}{W_{turbine}};$$

Methods to Improve Efficiency

1. Regeneration

Uses exhaust heat to preheat air

Improves efficiency

Saves fuel

2. Intercooling

Cooling between compressor stages

Reduces compressor work

Improves performance

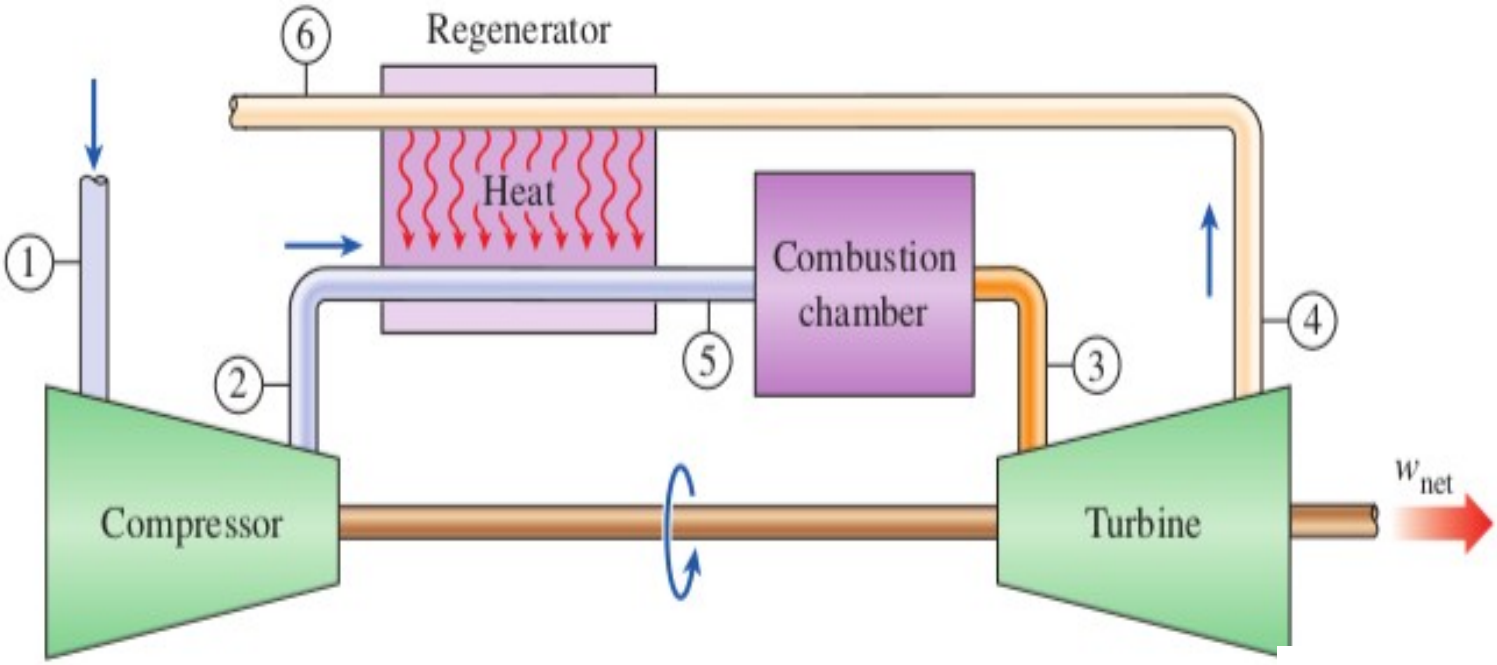
3. Reheating

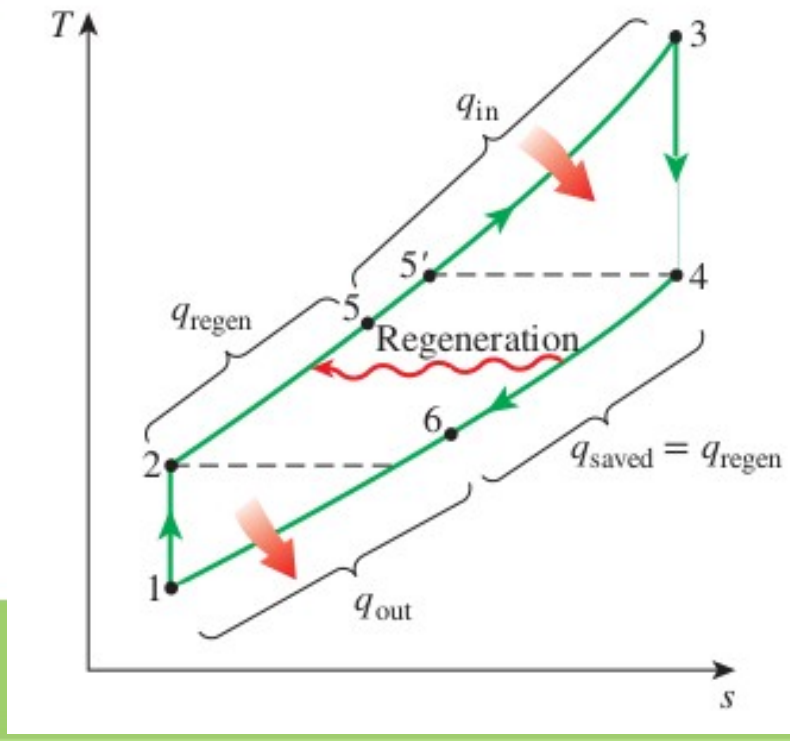
Expands gas in stages

Reheats between stages

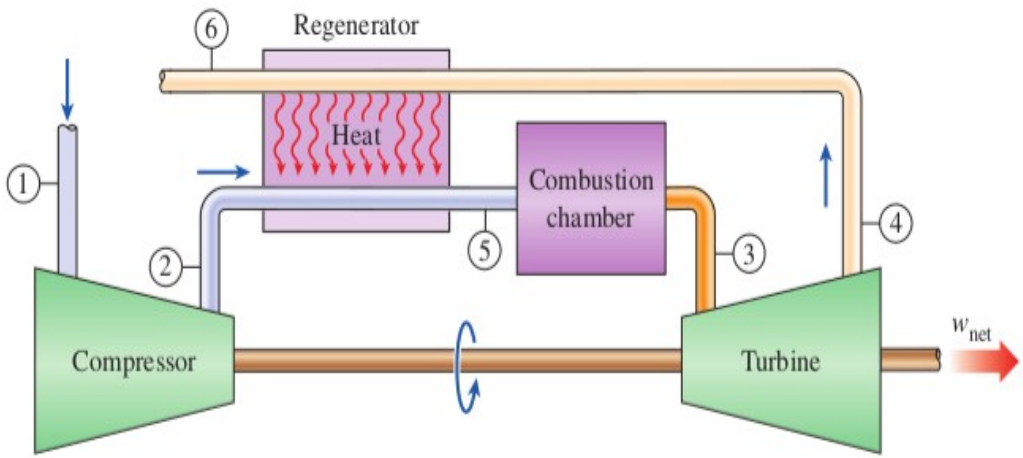
Increases output

Brayton Cycle With Regeneration



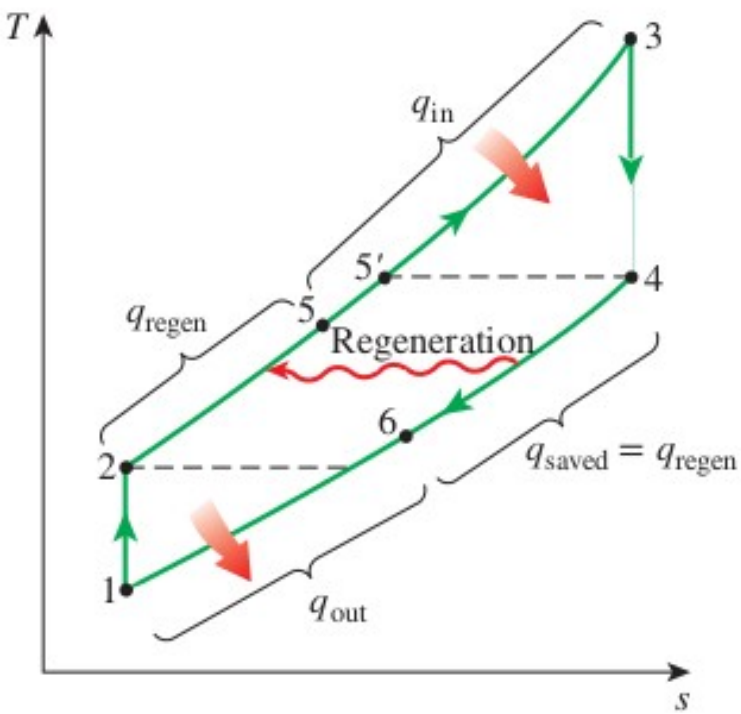


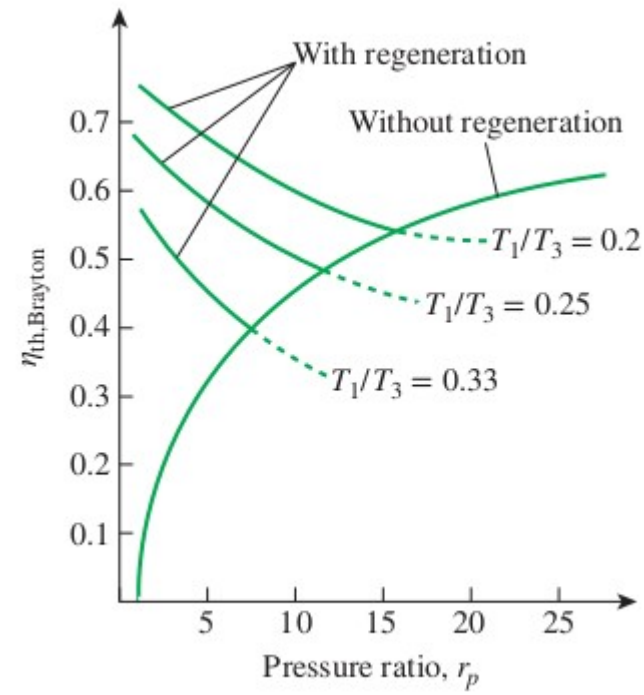
Brayton Cycle With Regeneration



$$\epsilon = \frac{q_{regen,act}}{q_{regen,max}}$$

$$\eta = 1 - \frac{T_1}{T_3} \cdot r_p^{\frac{\gamma-1}{\gamma}}$$

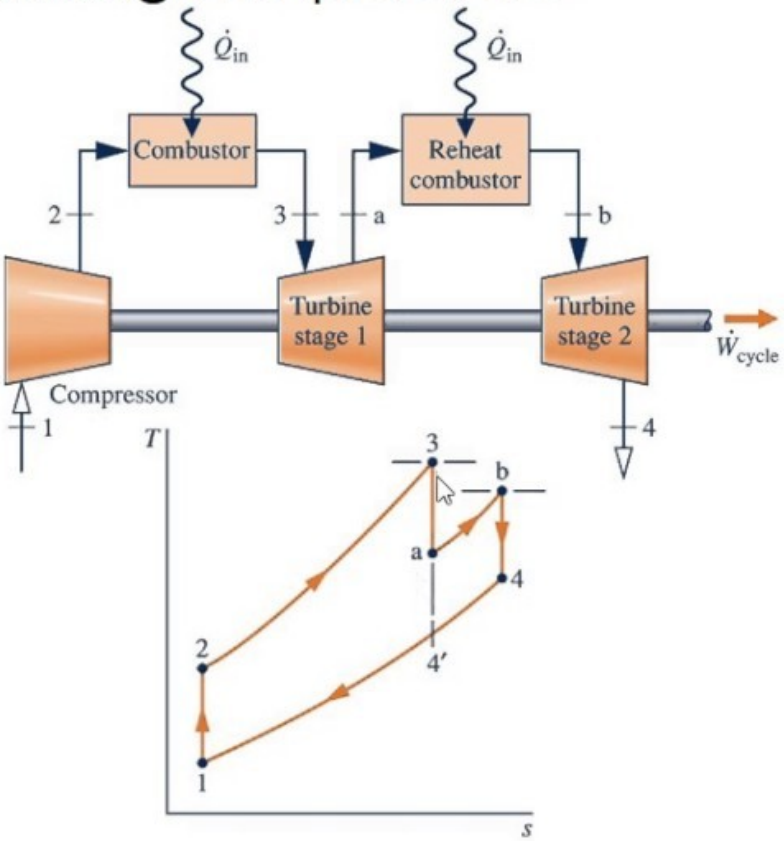




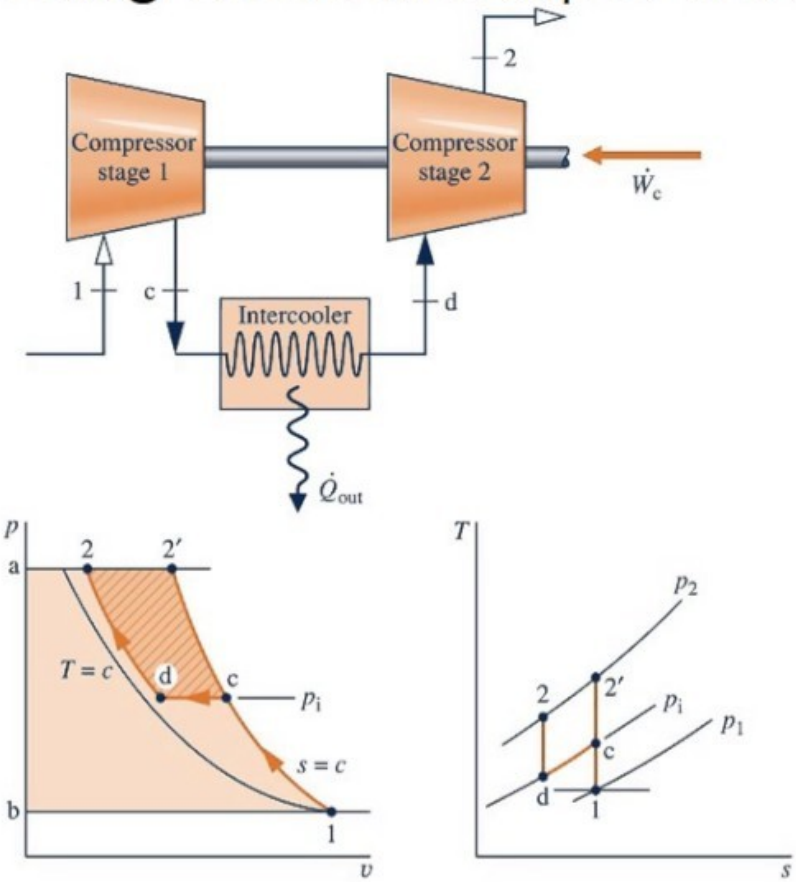
Brayton Cycle With Intercooling and Reheating

Gas turbines with reheat and intercooling

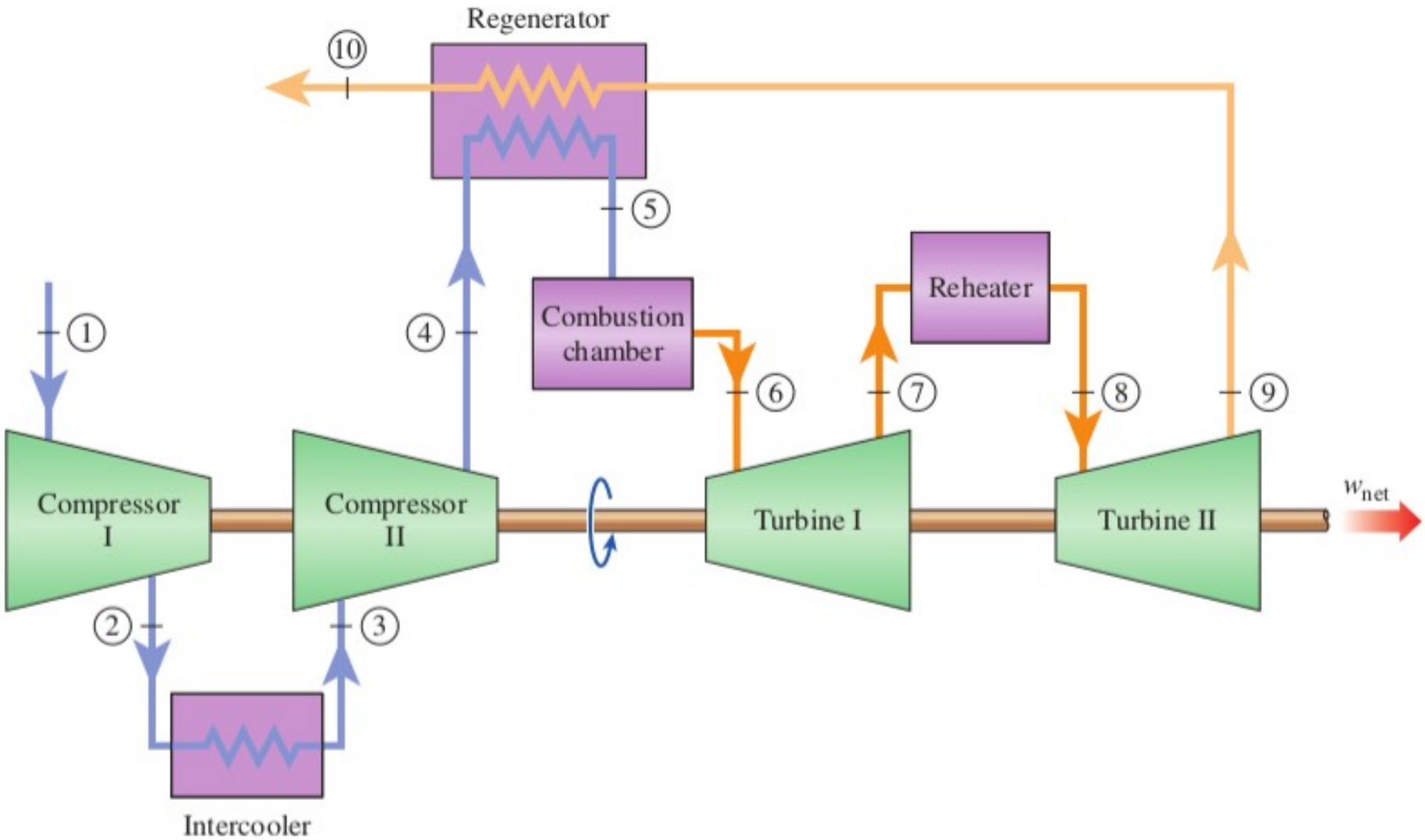
- Reheat increases work while limiting temperature



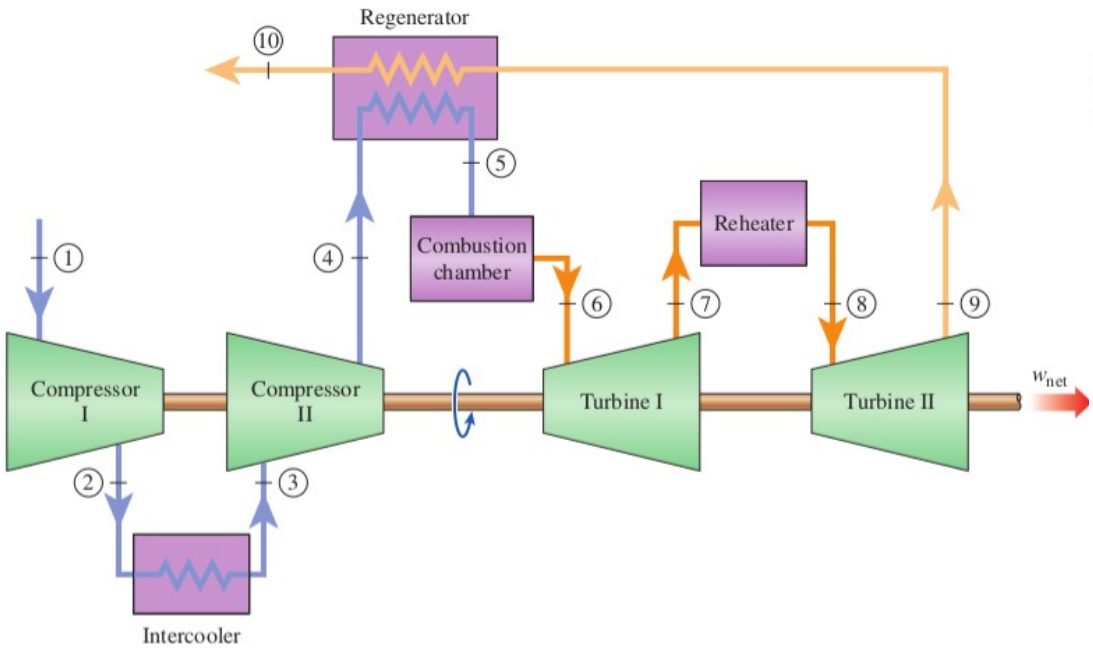
- Multistage compression with intercooling decreases input work

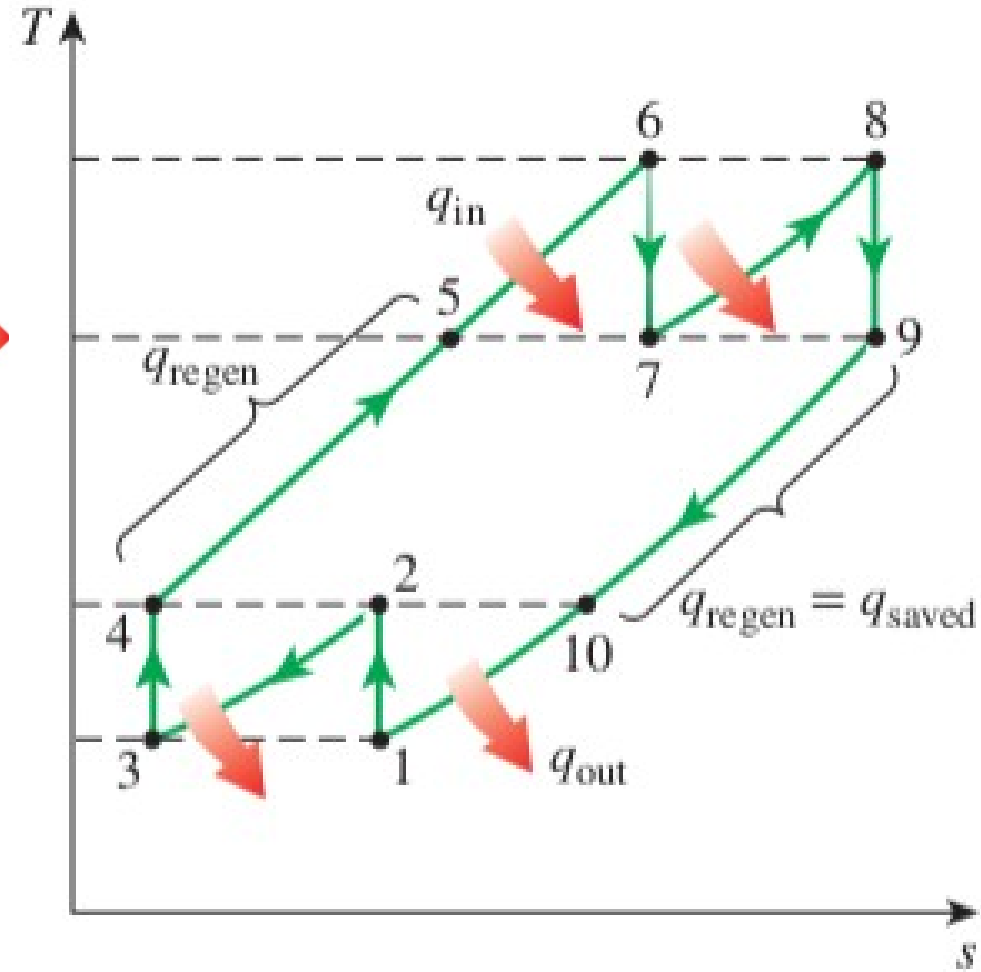


Brayton Cycle With Intercooling, Reheating and Regeneration



Brayton Cycle With Intercooling, Reheating and Regeneration





Fuels Used

Natural gas
Diesel oil
Kerosene
Producer gas

Gas Turbine Materials

Requirements:

High temperature resistance

Corrosion resistance

High strength

Materials Used:

Nickel alloys

Chromium alloys

Ceramic coatings

Combined Cycle Power Plant

Definition: Combination of gas turbine + steam turbine systems

Working:

Gas turbine produces power

Exhaust heat used to produce steam

Steam drives steam turbine

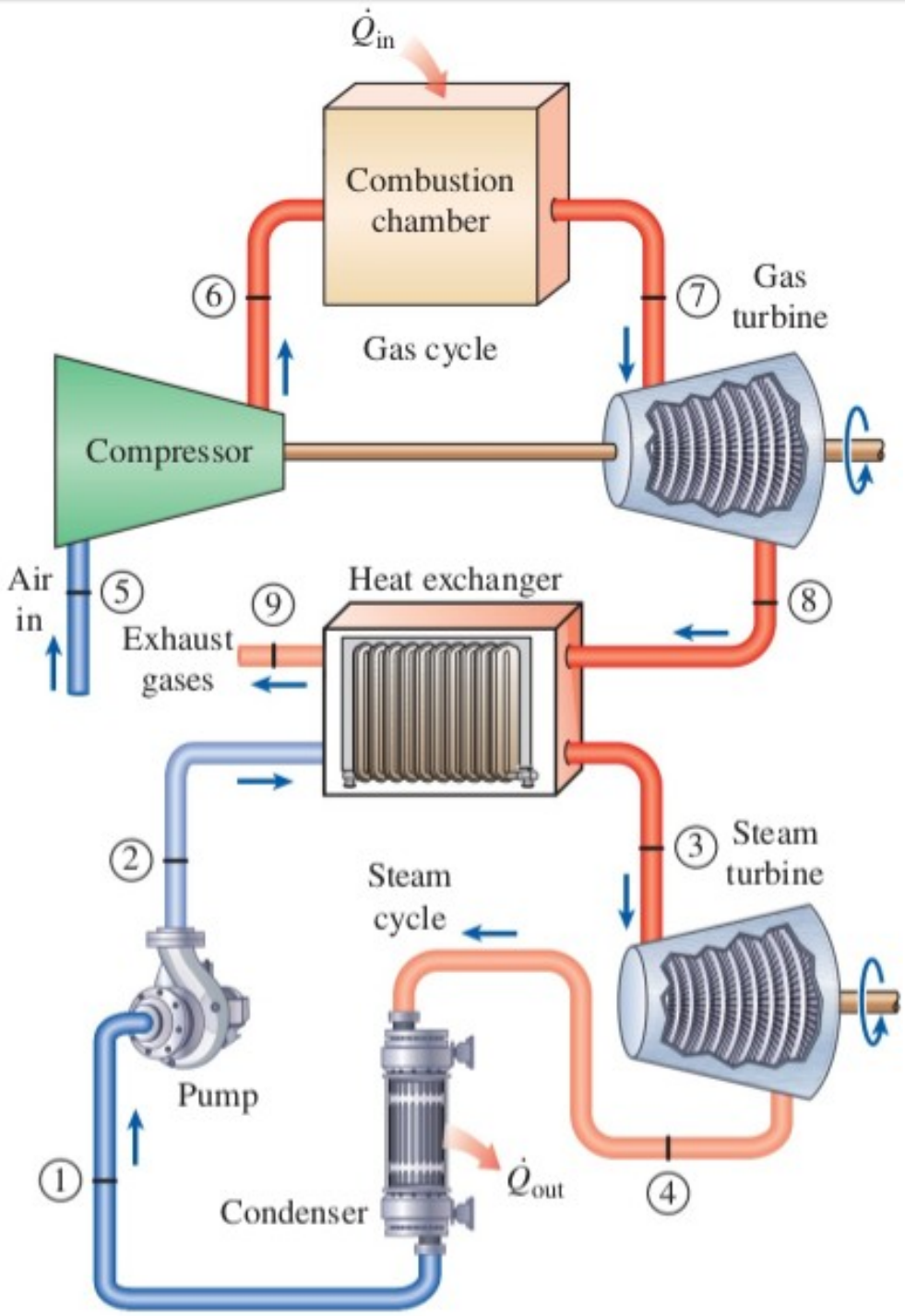
Advantages:

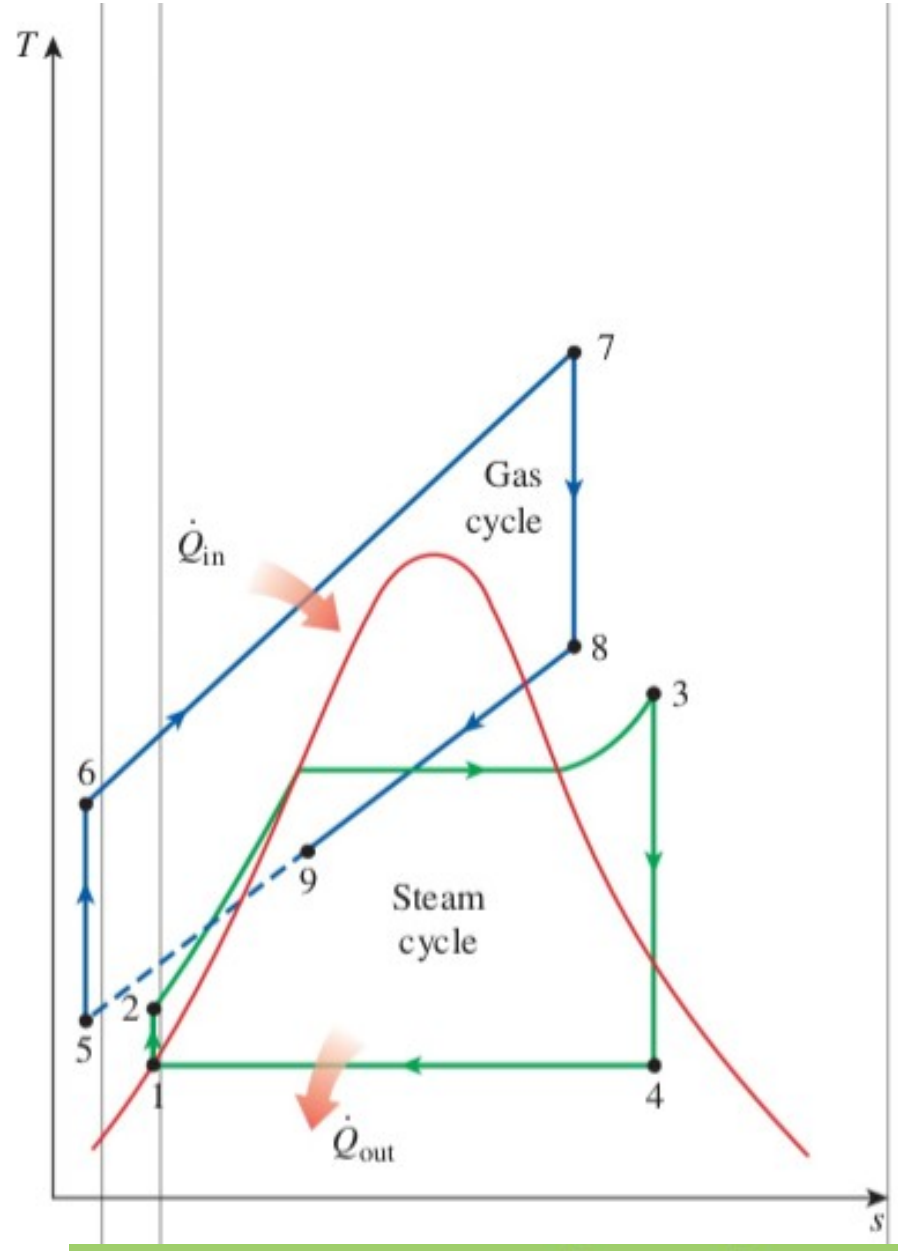
Very high efficiency (~60%)

Better fuel utilization

Reduced waste heat

Combined Cycle Power Plant





Applications of Gas Turbine Plant

Aircraft propulsion
Power generation
Marine engines
Industrial plants

Advantages

High power-to-weight ratio
Quick start
Compact size

Disadvantages

High initial cost

Requires skilled maintenance

Efficiency drops at part load